

Fundamental Matrix

Lecture-13

Transformations Between Two Images

- Translation
- Rotation
- Rigid
- Similarity (scaled rotation)
- Affine
- Projective
- Pseudo Perspective
- Bi-linear

Applications

- Stereo
- Structure from Motion
- View Invariant Action Recognition
- ..

Stereo Pairs and Depth Maps (from Szeliski's book)



(a)



(b)



(c)



(d)



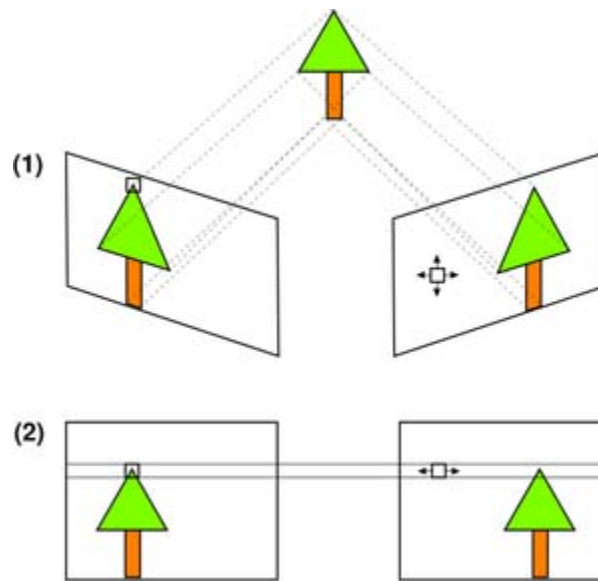
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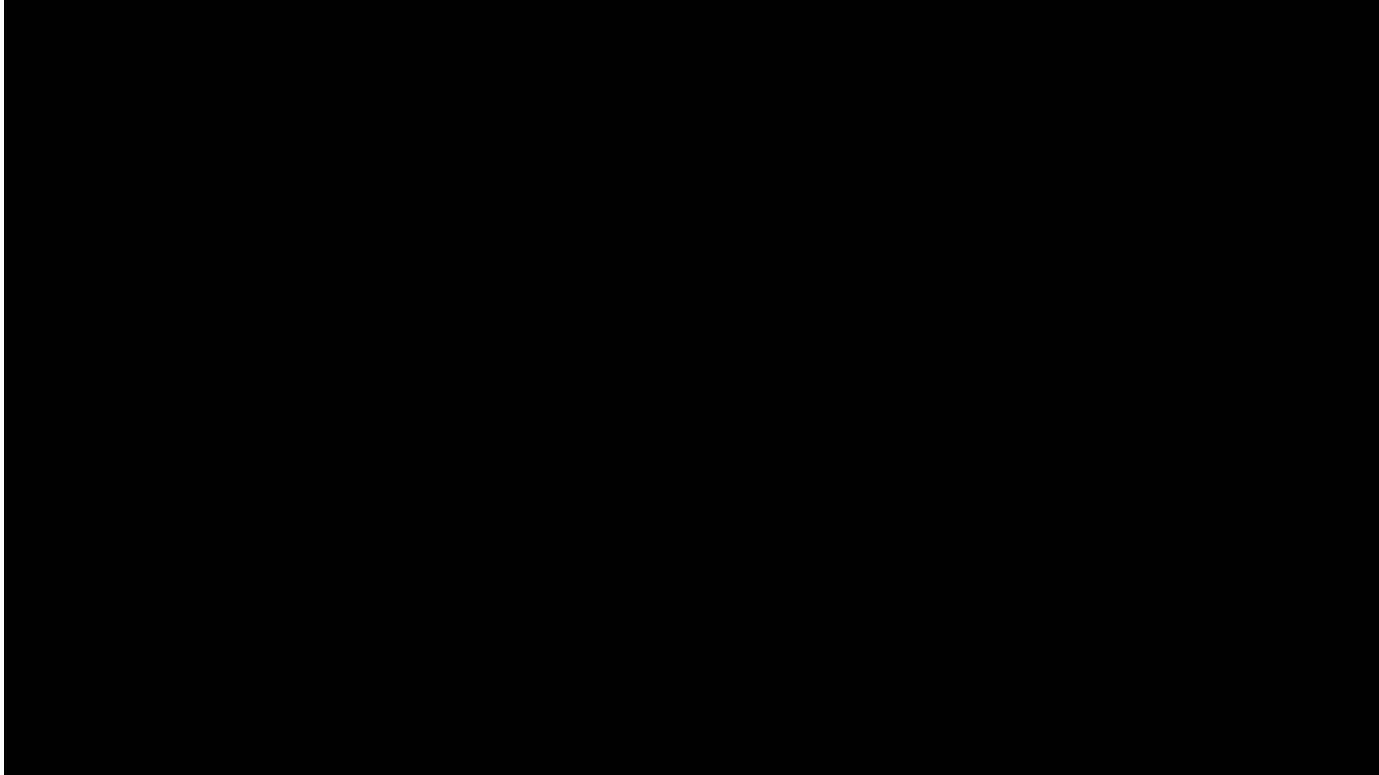
(f)



Image Rectification For Stereo



Photosynth



Fundamental Matrix

- Longuet Higgins (1981)
- Hartley (1992)
- Faugeras (1992)
- Zhang (1995)

Fundamental Matrix Song

<http://www.youtube.com/watch?v=DgGV3l82NTk>

Preliminaries

- Linear Independence
- Rank of a Matrix
- Matrix Norm
- Singular Value Decomposition
- Vector Cross product to Matrix Multiplication
- RANSAC

Linearly Independence

A finite subset of n vectors, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, from the vector space V , is ***linearly dependent*** if and only if there exists a set of n scalars, $a_1, a_2, \dots, a_n$, not all zero, such that

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = \mathbf{0}.$$

Rank of a Matrix

- The **column rank** of a matrix A is the maximum number of linearly independent column vectors of A .
- The **row rank** of a matrix A is the maximum number of linearly independent row vectors of A .
- The column rank of A is the dimension of the column space of A
- The row rank of A is the dimension of the row space of A .

Example (Row Echelon)

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix} R_2 \rightarrow 2R_1 + R_2$$

Rank is 2

Singular Value Decomposition (SVD)

Theorem: Any m by n matrix A , for which $m \geq n$, can be written as

$$A = O_1 \Sigma O_2$$

$m \times n \quad \quad m \times n \quad n \times n \quad n \times n$

Σ is diagonal

O_1, O_2 are orthogonal

$$O_1^T O_1 = O_2^T O_2 = I$$

Matrix Norm

L1 matrix norm is maximum of absolute column sum.

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|,$$

Vector Cross Product to Matrix-vector multiplication

$$A \times B = \begin{bmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{bmatrix} = \begin{bmatrix} -A_z B_y + A_y B_z & A_z B_x - A_x B_z & -B_x A_y + B_y A_x \end{bmatrix}$$

$$A \times B = S.B = \begin{bmatrix} 0 & -A_z & A_y \\ A_z & 0 & -A_x \\ -A_y & A_x & 0 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} -A_z B_y + A_y B_z & A_z B_x - A_x B_z & -B_x A_y + B_y A_x \end{bmatrix}$$

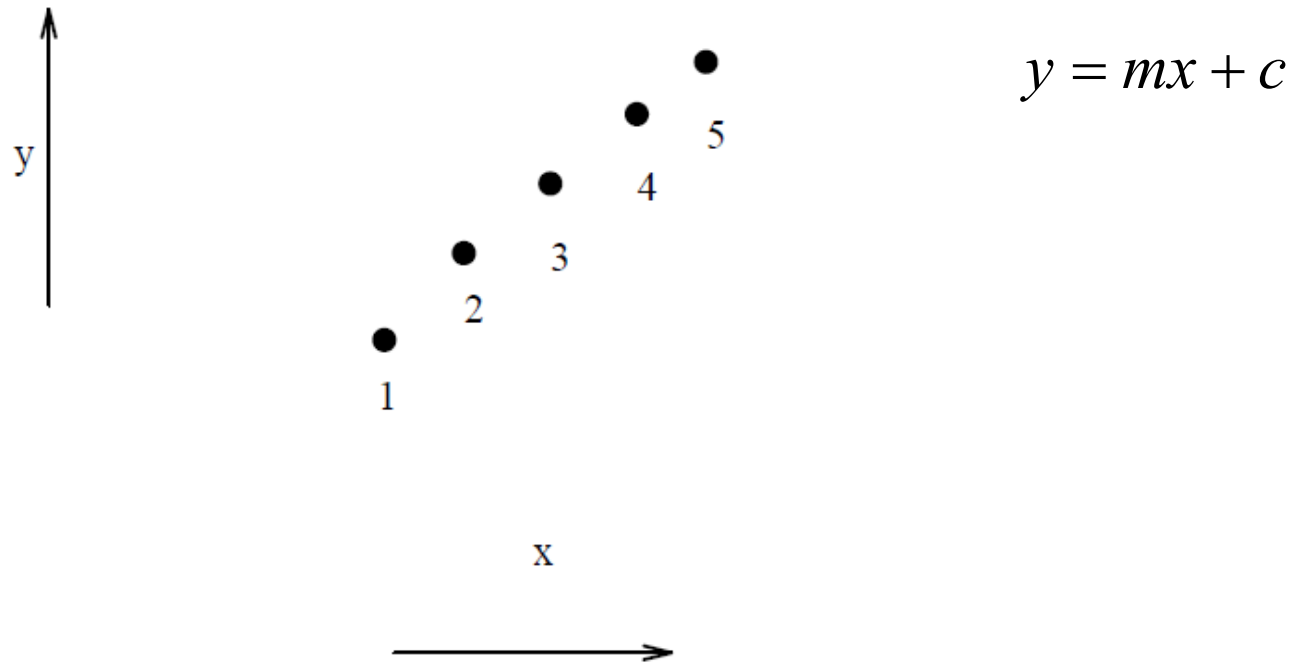
How to Fit A Line?

- Least squares Fit (over constraint)
- RANSAC (constraint)
- Hough Transform (under constraint)

RANSAC Song

<http://www.youtube.com/watch?v=1YNjMxxXO-E&feature=relmfu>

How to Fit A Line?



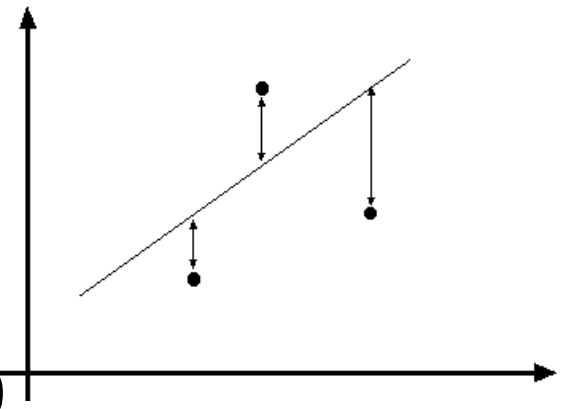
Least Squares Fit

- Standard linear solution to estimating unknowns.
 - If we know which points belong to which line
 - Or if there is only one line

$$y = mx + c = f(x, m, c)$$

$$\text{Minimize } E = \sum_i [y_i - f(x_i, m, c)]^2$$

Take derivative wrt m and c set to 0

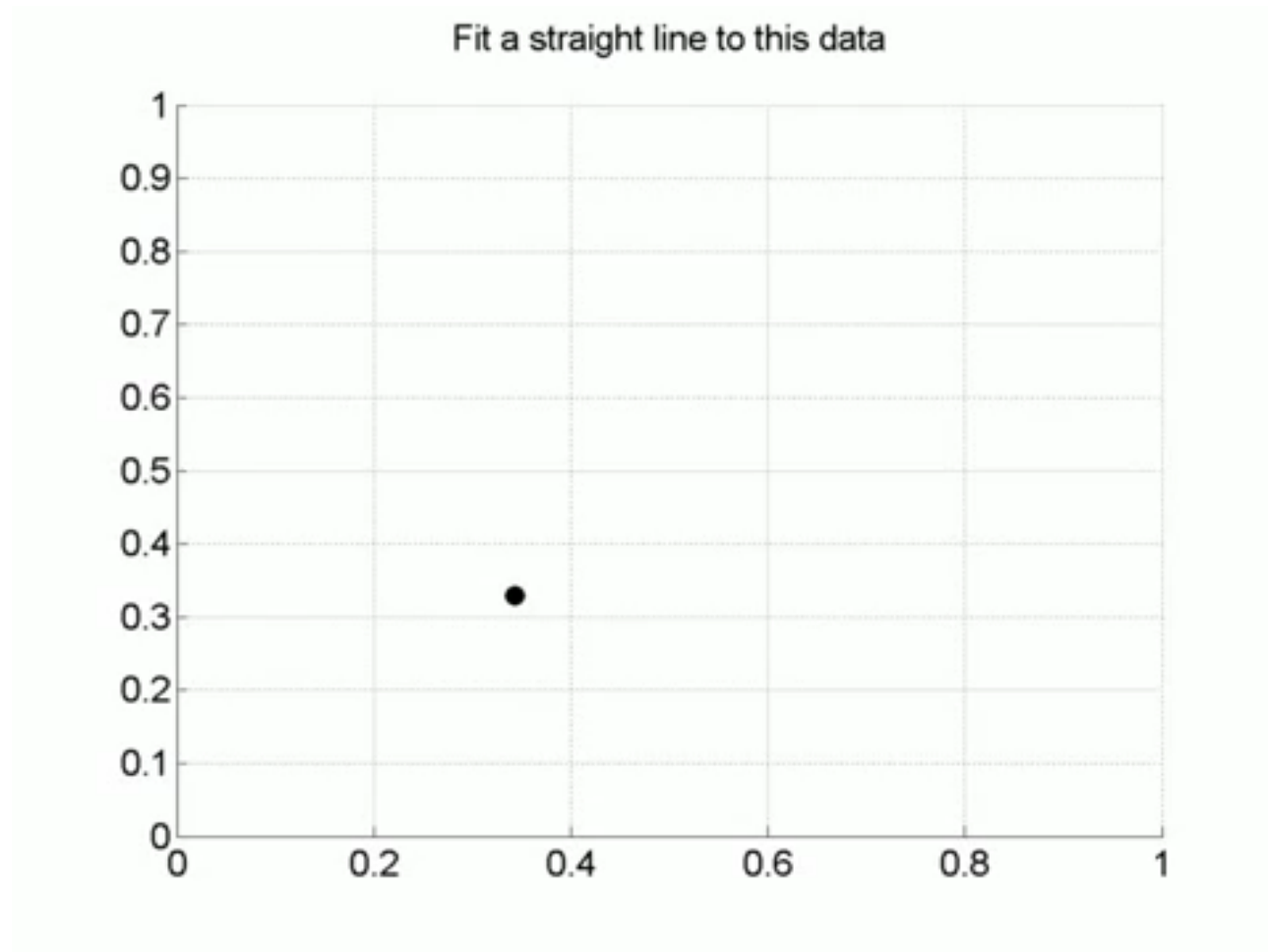


Line Fitting

$$\begin{array}{l} y = mx + c \\ \downarrow \\ \begin{array}{l} y_1 = mx_1 + c \\ y_2 = mx_2 + c \\ \vdots \\ y_n = mx_n + c \end{array} \end{array} \quad \nearrow \quad \begin{array}{l} \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_B = \underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & 1 \\ x_n & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} m \\ c \end{bmatrix}}_D \\ \downarrow \\ \begin{array}{l} A^T B = A^T A D \\ (A^T A)^{-1} A^T B = (A^T A)^{-1} (A^T A) D \\ D = (A^T A)^{-1} A^T B \end{array} \end{array} \Rightarrow B = AD$$

RANSAC: Random Sampling and Consensus

RANSAC Song



<http://www.youtube.com/watch?v=1YNjMxxXO-E&feature=relmfu>

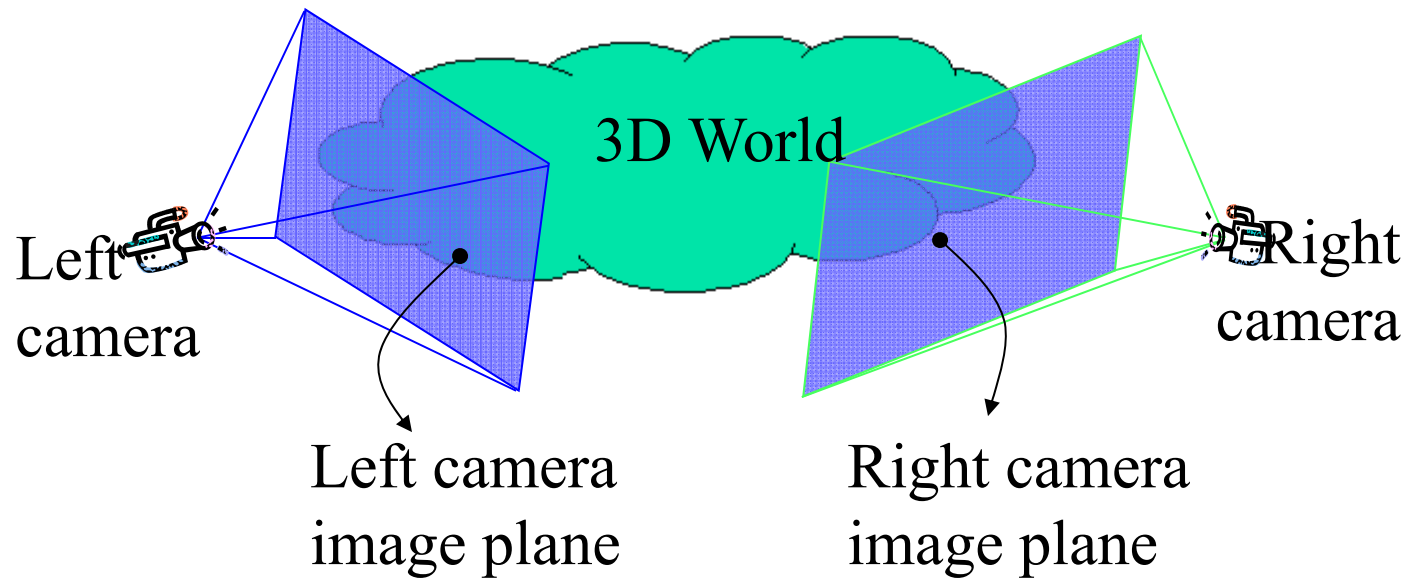
RANSAC: Random Sampling and Consensus

1. Randomly select two points to fit a line
2. Find the error between the estimated solution and all other points. If the error is less than tolerance, then quit, else go to step (1).

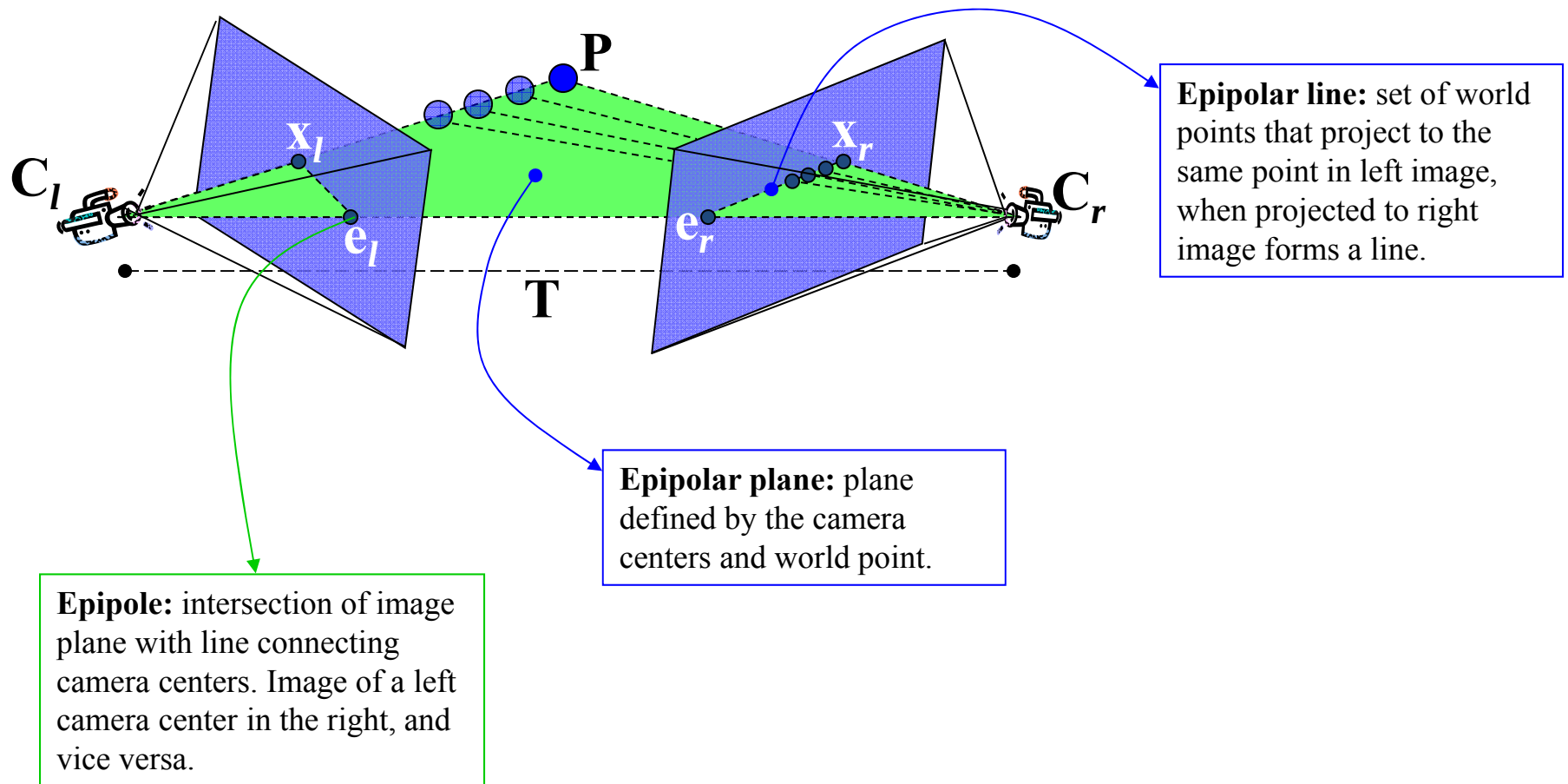
Derivation of Fundamental Matrix

Epipolar Geometry

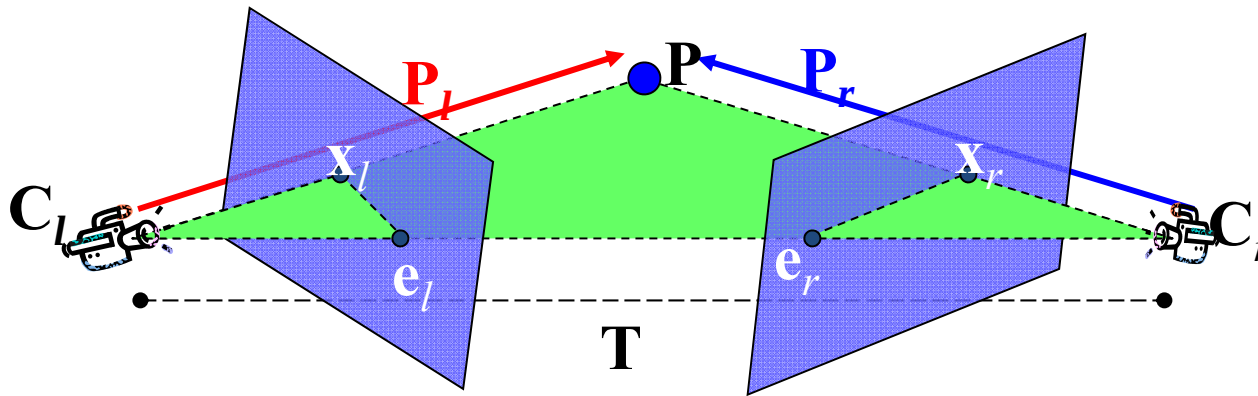
- Defined for two static cameras



Epipolar Geometry



Essential Matrix



Coplanarity constraint between vectors $(\mathbf{P}_l - \mathbf{T})$, $\mathbf{T}$, $\mathbf{P}_l$.

$$\left. \begin{array}{l} (\mathbf{P}_l - \mathbf{T})^T \mathbf{T} \times \mathbf{P}_l = 0 \\ \mathbf{P}_r = \mathbf{R}(\mathbf{P}_l - \mathbf{T}) \end{array} \right\} \mathbf{P}_r^T \mathbf{R} \mathbf{T} \times \mathbf{P}_l = 0$$

$$\mathbf{R}^T \mathbf{P}_r = (\mathbf{P}_l - \mathbf{T})$$

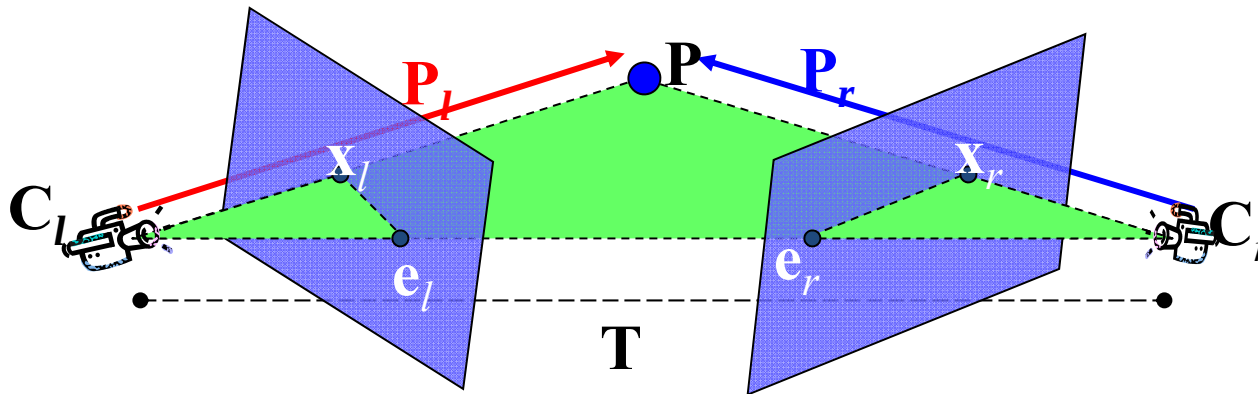
$$\mathbf{P}_r^T \mathbf{R} = (\mathbf{P}_l - \mathbf{T})^T$$

Vector Cross Product to Matrix-vector multiplication

$$A \times B = \begin{bmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{bmatrix} = \begin{bmatrix} -A_z B_y + A_y B_z & A_z B_x - A_x B_z & -B_x A_y + B_y A_x \end{bmatrix}$$

$$A \times B = S.B = \begin{bmatrix} 0 & -A_z & A_y \\ A_z & 0 & -A_x \\ -A_y & A_x & 0 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} -A_z B_y + A_y B_z & A_z B_x - A_x B_z & -B_x A_y + B_y A_x \end{bmatrix}$$

Essential Matrix



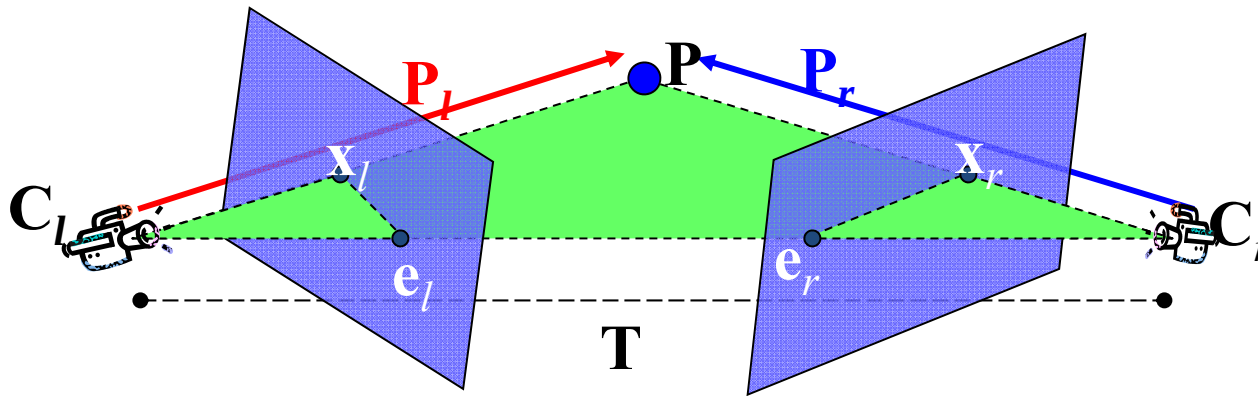
$$\left. \mathbf{P}_r^T \mathbf{R} \mathbf{T} \times \mathbf{P}_l = 0 \right\} \mathbf{P}_r^T \mathbf{R} \boxed{\mathbf{S}} \mathbf{P}_l = 0 \quad \longrightarrow \quad \boxed{\mathbf{P}_r^T \mathbf{E} \mathbf{P}_l = 0}$$

essential matrix

$$\mathbf{E} = \mathbf{R} \mathbf{S}$$

$$\begin{pmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{pmatrix}$$

Fundamental Matrix



Apply Camera model

$$M_l^{-1} \mathbf{x}_l = \mathbf{P}_l$$

$$M_r^{-1} \mathbf{x}_r = \mathbf{P}_r$$

$$\mathbf{x}_r^T M_r^{-T} = \mathbf{P}_r^T$$

$$\mathbf{x}_l = M_l \mathbf{P}_l$$

$$\mathbf{x}_r = M_r \mathbf{P}_r$$

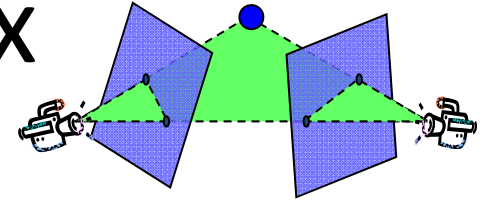
$$\mathbf{P}_r^T E \mathbf{P}_l = 0$$

$$\left. \begin{array}{l} \mathbf{x}_r^T M_r^{-T} E M_l^{-1} \mathbf{x}_l = 0 \\ \mathbf{x}_r^T (M_r^{-T} E M_l^{-1}) \mathbf{x}_l = 0 \end{array} \right\}$$

$$\boxed{\mathbf{x}_r^T F \mathbf{x}_l = 0}$$

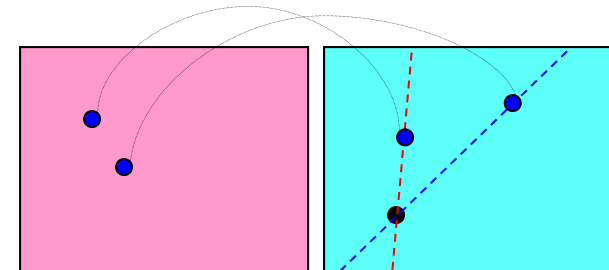
fundamental matrix

Fundamental Matrix

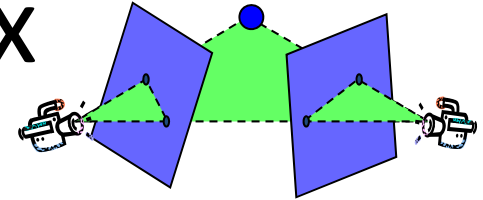


$$\mathbf{x}'^T F \mathbf{x} = \mathbf{x}'^T \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & m \end{pmatrix} \mathbf{x} = 0$$

- Given a point $\mathbf{x}$ in left camera, epipolar line in right camera is: $\mathbf{u}_r = F \mathbf{x}$



Fundamental Matrix



$$\mathbf{x}_l^T F \mathbf{x}_r = \mathbf{x}_l^T \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & m \end{pmatrix} \mathbf{x}_r = 0$$

- 3x3 matrix with 9 components
- Rank 2 matrix (due to S)
- 7 degrees of freedom
- Given a point in left camera $\mathbf{x}$, epipolar line in right camera is: $\mathbf{u}_r = F\mathbf{x}$

Fundamental Matrix

- Longuet Higgins (1981)
- Hartley (1992)
- Faugeras (1992)
- Zhang (1995)

Fundamental Matrix

- Fundamental matrix captures the relationship between the corresponding points in two views.

$$\begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}^T \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} = 0,$$

$$\begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}^T \begin{bmatrix} f_{11}x'_i + f_{12}y'_i + f_{13} \\ f_{21}x'_i + f_{22}y'_i + f_{23} \\ f_{31}x'_i + f_{32}y'_i + f_{33} \end{bmatrix} = 0,$$

Fundamental Matrix

$$x_i(f_{11}x' + f_{12}y'_i + f_{13}) + y_i(f_{21}x' + f_{22}y'_i + f_{23}) + (f_{31}x' + f_{32}y'_i + f_{33}) = 0$$

$$x_i x' f_{11} + x_i y'_i f_{12} + x_i f_{13} + y_i x' f_{21} + x' y'_i f_{22} + y'_i f_{23} + x' f_{31} + y'_i f_{32} + f_{33} = 0$$

One equation for one point correspondence

$$Mf = \begin{bmatrix} x'_1 x_1 & x'_1 y_1 & x'_1 & y'_1 x_1 & y'_1 y_1 & y'_1 & x_1 & y_1 & 1 \\ x'_2 x_2 & x'_2 y_2 & x'_2 & y'_2 x_2 & y'_2 y_2 & y'_2 & x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x'_n x_n & x'_n y_n & x'_n & y'_n x_n & y'_n y_n & y'_n & x_n & y_n & 1 \end{bmatrix} f = 0$$

M is 9 by n matrix $f = [f_{11} \quad f_{12} \quad f_{13} \quad f_{21} \quad f_{22} \quad f_{23} \quad f_{31} \quad f_{32} \quad f_{33}]$

To solve the equation, the rank(M) must be 8.

Computation of Fundamental Matrix

Normalized 8-point algorithm (Hartley)

Objective:

Compute fundamental matrix F such that $\mathbf{x}_i' F \mathbf{x}_i = 0$

Algorithm

Normalize the image

$$\hat{\mathbf{x}}_i = T \mathbf{x}_i$$

$$\hat{\mathbf{x}}_i' = T' \mathbf{x}_i'$$

$$T = \begin{bmatrix} a_x & 0 & d_x \\ 0 & a_y & d_y \\ 0 & 0 & 1 \end{bmatrix}$$

Find centroid of points in each image, determine the range, and normalize all points between 0 and 1

Linear solution

determining the eigen vector corresponding to the smallest eigen value of A ,

$$A f = \begin{bmatrix} \hat{x}_1' \hat{x}_1 & \hat{x}_1' \hat{y}_1 & \hat{x}_1' & \hat{y}_1' \hat{x}_1 & \hat{y}_1' \hat{y}_1 & \hat{y}_1' & \hat{x}_1 & \hat{y}_1 & 1 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \hat{x}_8' \hat{x}_8 & \hat{x}_8' \hat{y}_8 & \hat{x}_8' & \hat{y}_8' \hat{x}_8 & \hat{y}_8' \hat{y}_8 & \hat{y}_8' & \hat{x}_8 & \hat{y}_8 & 1 \end{bmatrix} f = 0$$

Normalized 8-point algorithm (Hartley)

Construct

$$\hat{F} = \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix}$$

Normalize

$$\hat{\hat{F}} = \hat{F} / \|\hat{F}\|$$

L1 matrix norm is maximum of absolute column sum.

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|,$$

Constraint enforcement SVD decomposition

$$\hat{\hat{F}} = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} V' \quad (\sigma_1 \geq \sigma_2 \geq \sigma_3)$$

Rank enforcement

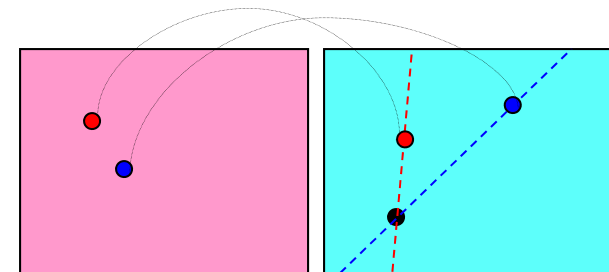
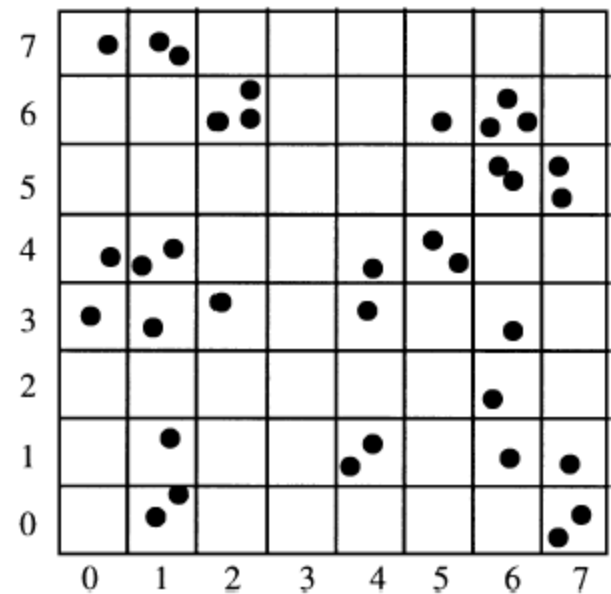
$$\tilde{F} = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} V' \quad (\sigma_3 = 0)$$

De-normalization:

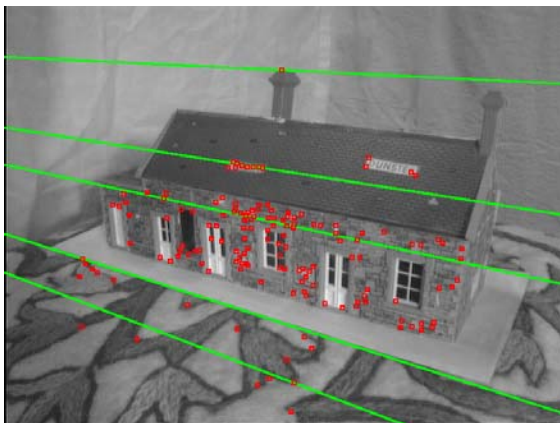
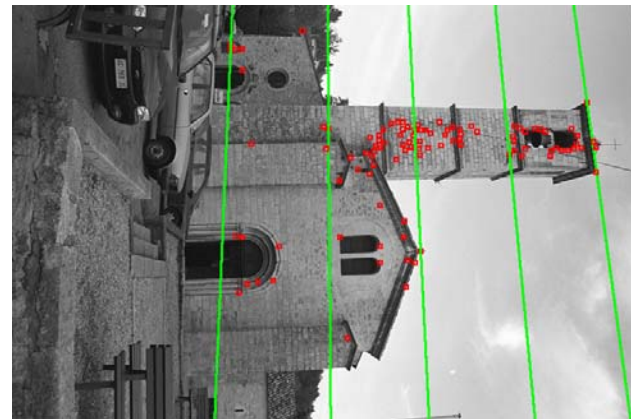
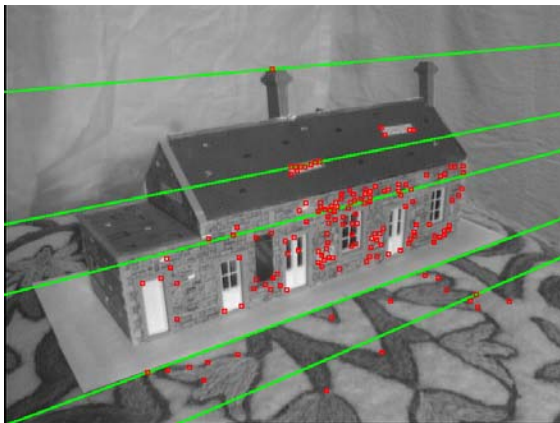
$$F = T'^T \tilde{F} T$$

Robust Fundamental Matrix Estimation (by Zhang)

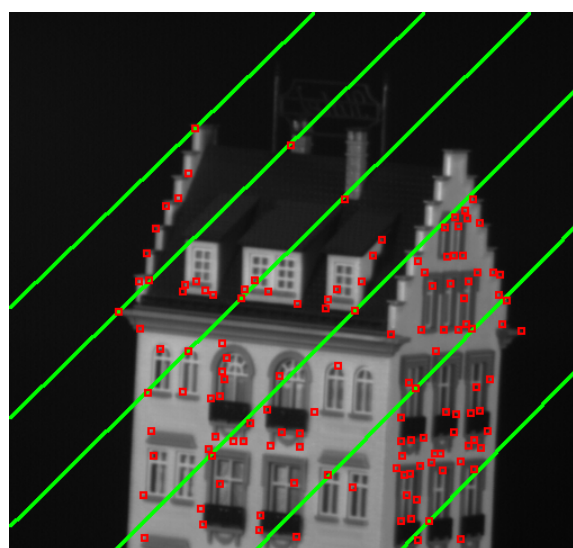
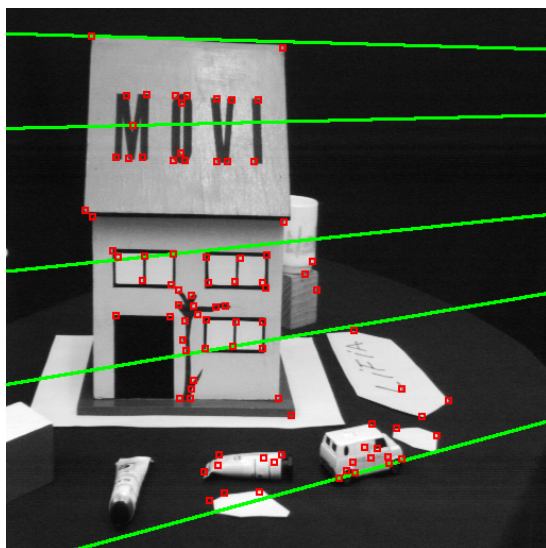
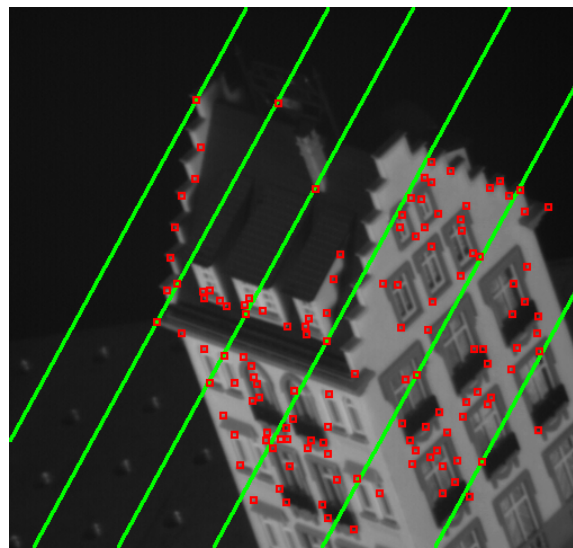
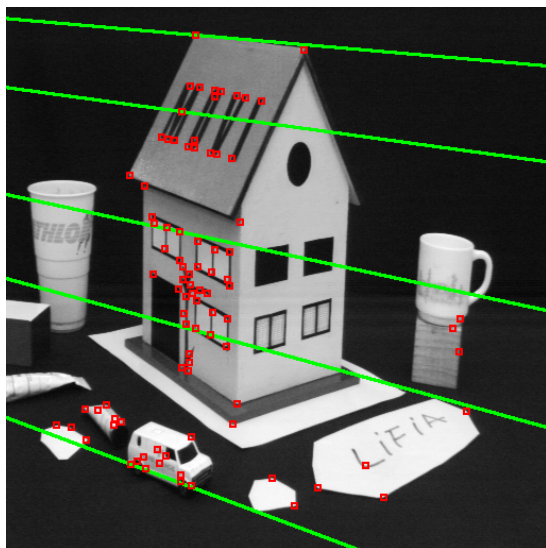
- Uniformly divide the image into 8×8 grid.
- Randomly select 8 grid cells and pick one pair of corresponding points from each grid cell, then use Hartley's 8-point algorithm to compute Fundamental Matrix F_i .
- For each F_i , compute the median of the squared residuals R_i .
 - $R_i = \text{median}_k [d(p_{1k}, F_i p_{2k}) + d(p_{2k}, F_i' p_{1k})]$
- Select the best F_i according to R_i .
- Determine outliers if $R_k > Th$.
- Using the remaining points compute the fundamental Matrix F by weighted least square method.



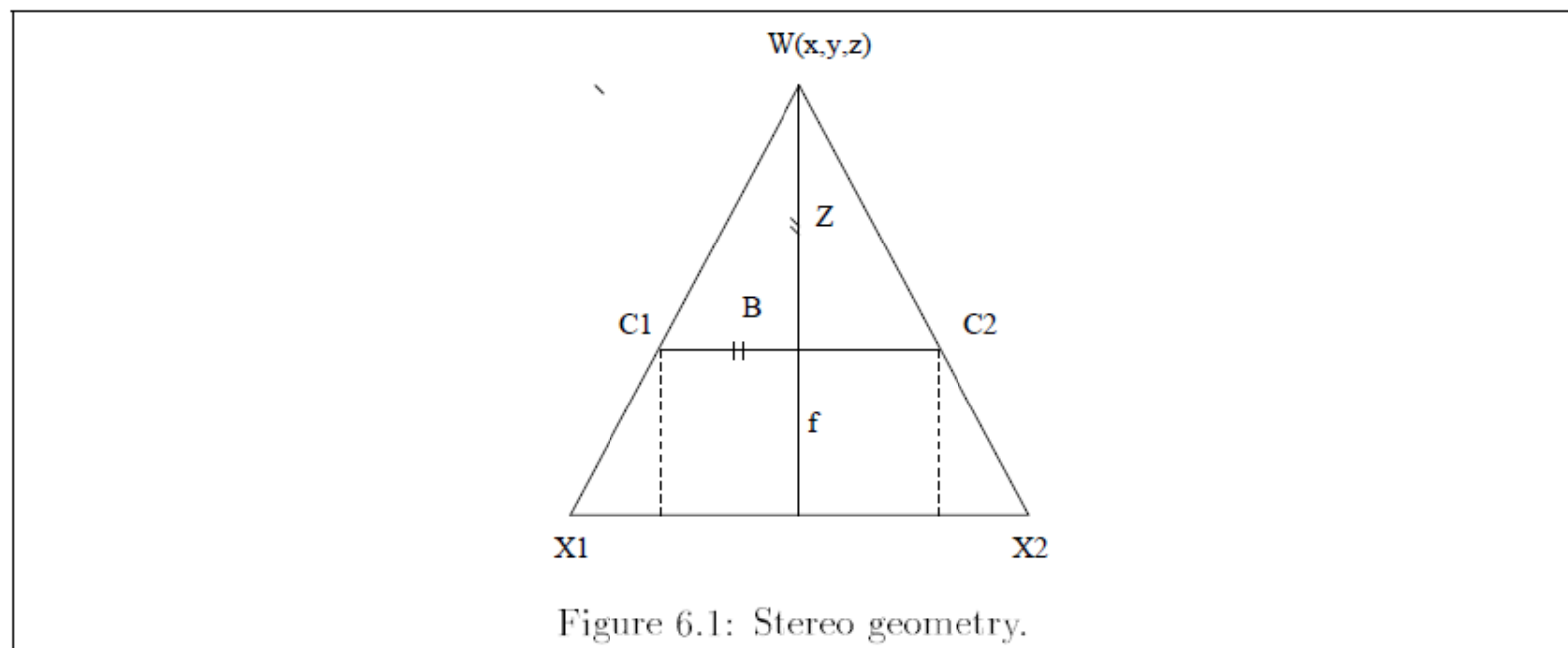
Epi-polar Lines



Epi-polar lines



Stereo



$$\frac{Z + f}{Z} = \frac{x_1 + x_2 + B}{B},$$

$$Z = \frac{fB}{x_1 + x_2},$$

Stereo Pairs and Depth Maps (from Szeliski's book)



(a)



(b)



(c)



(d)



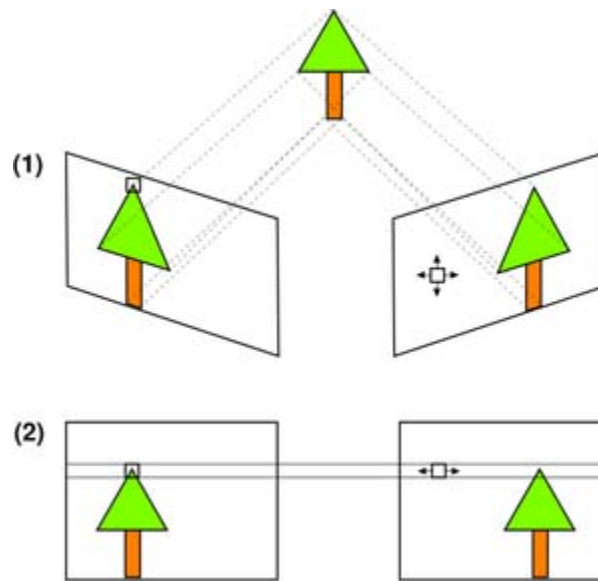
(e)



(f)



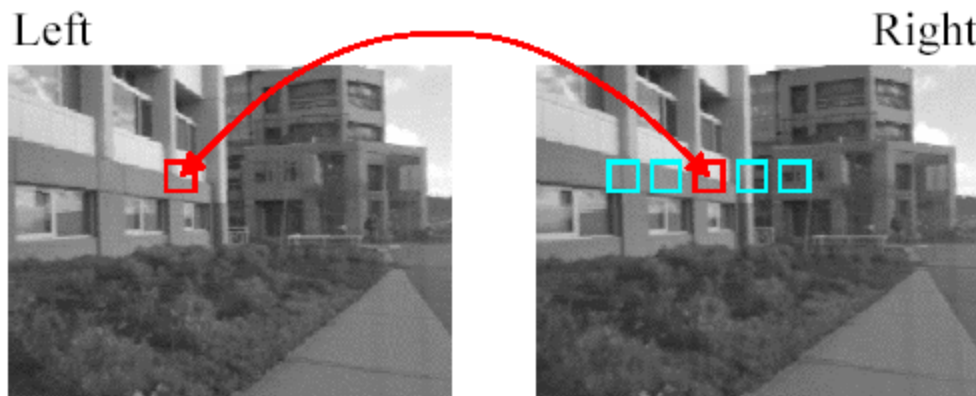
Image Rectification For Stereo



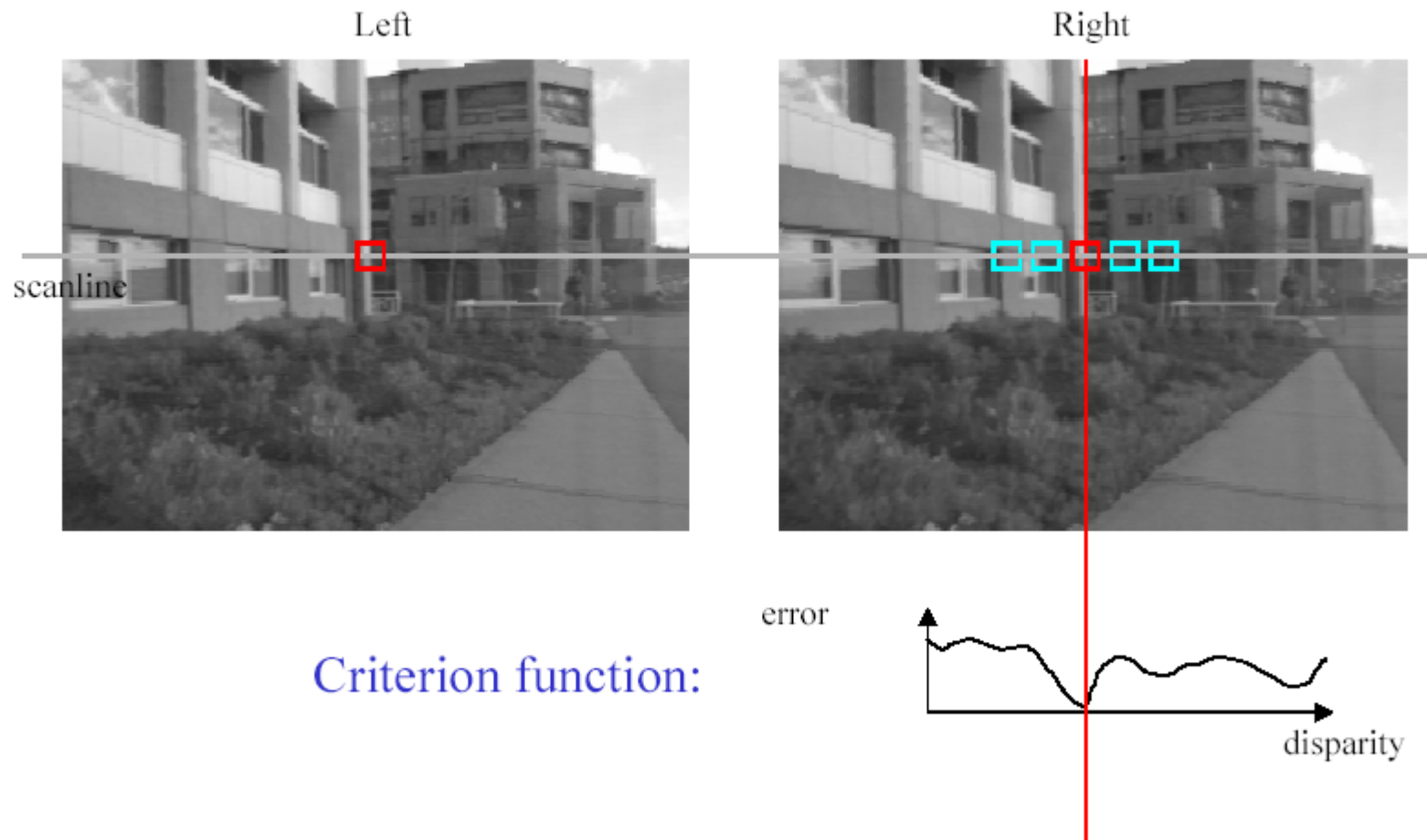
Correlation Based Stereo Methods

- Once disparity is available compute depth using

$$Z = \frac{fB}{d}$$



Correspondence using Search



Correlation Based Stereo Methods

- Depth is computed only at tokens and interpolated/extrapolated to remaining pixel
- Disparity map is constructed based on a correlation measure

$$\text{SSD} = \sum \sum (I_{\text{left}} - I_{\text{right}})^2$$

$$\text{AD} = \sum \sum |I_{\text{left}} - I_{\text{right}}|$$

$$\text{CC} = \sum \sum I_{t+1} I_t$$

$$\text{NC} = \frac{\sum \sum (I_{\text{left}} \cdot I_{\text{right}})}{\sqrt{\sum \sum I_{\text{left}} \cdot I_{\text{right}}}}$$

$$\text{MC} = \frac{1}{64\sigma_{\text{left}}\sigma_{\text{right}}} \sum \sum (I_{\text{left}} - \mu_{\text{right}})(I_{\text{left}} - \mu_{\text{right}})$$



Barnard's Stereo Method

- Similar intensity
 - Similar to brightness constraint
- Smoothness of disparity

$$E = \sum_{i=-1}^1 \sum_{j=-1}^1 \|I_{left}(x+i, y+j) - I_{right}(x+i + D_x(x, y), y+j)\| + \lambda \|\nabla D(x, y)\|$$

$$\nabla D(x, y) = \sum_{i=-1}^1 \sum_{j=-1}^1 |D(x+i, y+j) - D(x, y)|$$



Barnard's Stereo Method

- Energy can be minimized using brute force search
 - Let max allowed disparity is 10 pixels
 - For 128x128 image for 10 possible levels of disparity
 - There 10^{16384} possible disparity values
 - We can select any minimization technique
 - Barnard choose simulated annealing

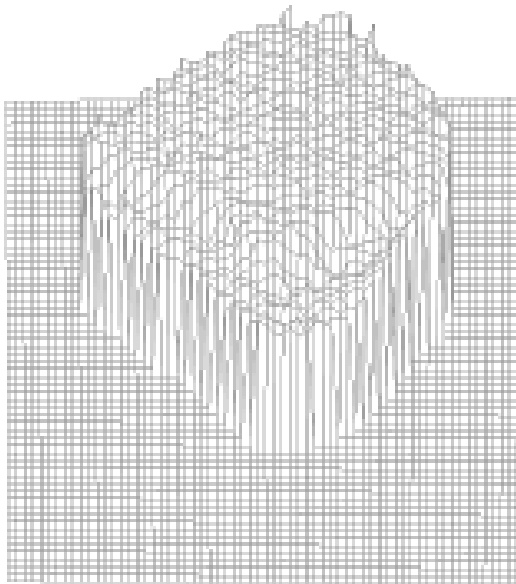


Simulated Annealing

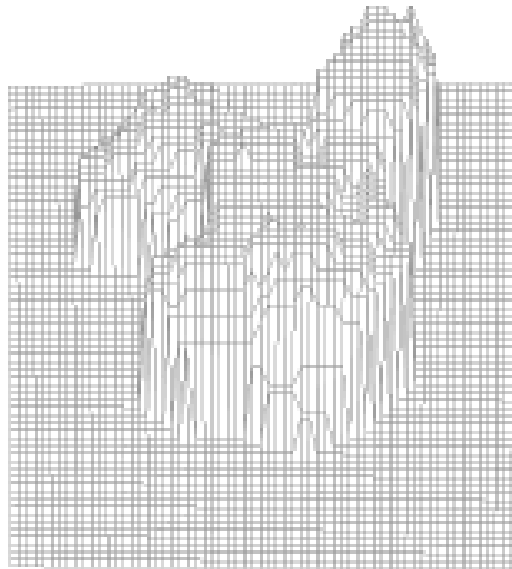
- Select a random state S (disparities)
- Select a high temperature
 - Select random S'
 - Compute $\Delta E = E(S') - E(S)$
 - If $(\Delta E < 0)$ $S \leftarrow S'$
 - Else
 - $P \leftarrow \exp(-\Delta E/T)$
 - $X \leftarrow \text{random}(0,1)$
 - If $X < P$ then $S \leftarrow S'$
 - If no decrease in several iterations lower T



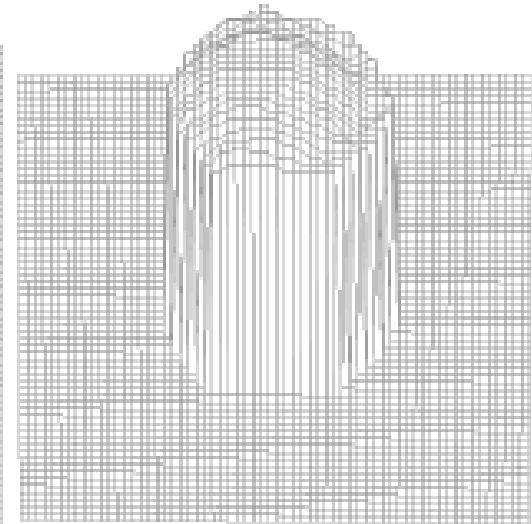
Examples



bread



toy

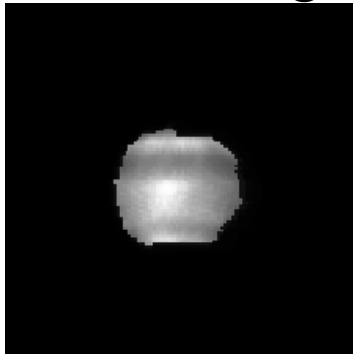


apple

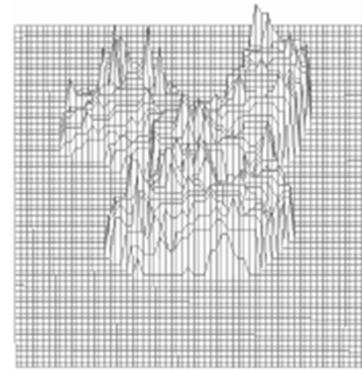
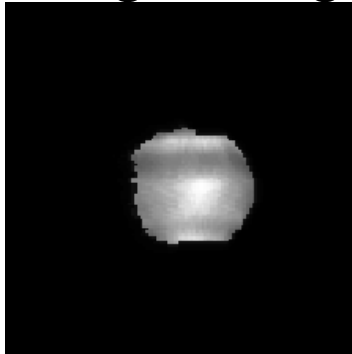
Examples



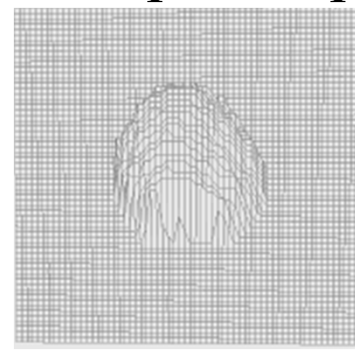
Left Image



Right Image



Depth Map



Stereo results

- Data from University of Tsukuba
- Similar results on other images without ground truth

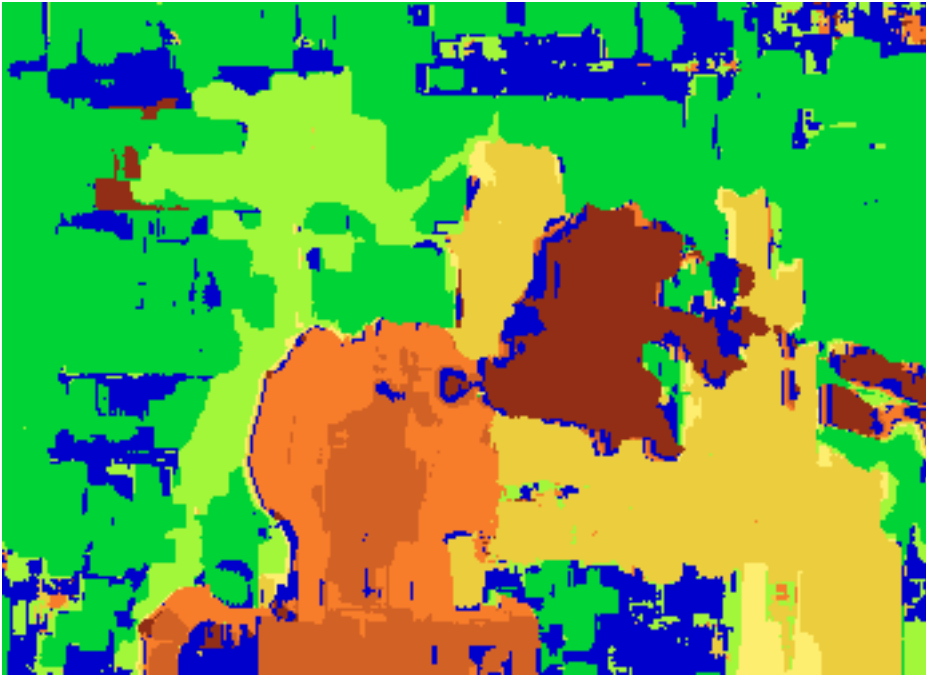


Scene



Ground truth

Results with window correlation



Window-based matching
(best window size)



Ground truth

Results with better method



State of the art method

Boykov et al., [Fast Approximate Energy Minimization via Graph Cuts](#),
International Conference on Computer Vision, September 1999.



Ground truth

Applications of Stereo (from Szeliski's book)



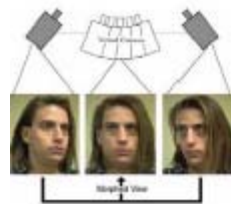
(a)



(b)



(c)



(d)



(e)



(f)



(g)



(h)



(i)



(j)

Reading Material

- Fundamental of Computer Vision
 - 6.2.1, 6.2.4 and 6.2.5
- Computer Vision: Algorithms and Applications, Richard Szeliski
 - Chapter 11